

RNA Isolation Procedure from Flow Sorted Cells

Source: Michelle Southard-Smith Laboratory

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This protocol was optimized from procedures shared by Sean Morrison's group (University of Michigan, primarily Jack Mosher), Steve Potter's group (Cincinnati Children's Hospital Medical Center) and David Miller's group (Vanderbilt University Medical Center) on RNA isolation from flow-sorted cell populations. Methods were optimized by Michelle Southard-Smith, Stephanie Byers-Skelton and Dennis Buehler.

Reagents used for this procedure:

TRIzol-LS, Invitrogen (Cat# 10296-010)
Ambion Glycogen, Invitrogen (Cat# AM9510)
RNeasy Micro Kit, Qiagen (Cat# 74004)
Isopropanol, OmniSolv (Cat# PX1834-6)
DNase I, Ambion (Cat# AM2222)
SUPERase•In, Ambion (Cat# AM2694)
Nuclease Free H₂O, Ambion (Cat# AM9932)
B-Mercaptoethanol (BME), Sigma (Cat# M7522)
Chloroform, Acros (Cat# 15821-0010)
Phase Lock Gel Tubes – Heavy 2ml, 5Prime (Cat# 2302830)
RNaseZap, ThermoFisher Scientific (Cat# AM9780)

Procedure:

1. FACs sort cells directly into 0.75 ml of TRIzol-LS, vortexing immediately after sort. After flow sort adjust volume up to 1 ml using Ambion Nuclease Free H₂O. Do not exceed 1 ml total volume.
2. Add 50 µl Glycogen to the TRIzol-LS/cell lysate. Vortex thoroughly!
3. Add 200 µl Chloroform and vortex thoroughly, then shake for 15 seconds.
4. Transfer into Phase Lock Gel Tubes immediately
5. Let mixture sit for 5 minutes at room temperature so phases begin to separate.
6. Centrifuge for 10 minutes at 4°C, 12,200 RCF
7. Remove 550 µl of upper clear aqueous and add directly into 550 µl of cold isopropanol.
8. Invert tube to mix and let sit on ice for 15 minutes.
9. Centrifuge to pellet RNA for 15 minutes at 4°C, 12,200 RCF.
10. After centrifugation small pellet should be visible at bottom of tube. Aspirate off the aqueous with a 1 ml pipette tip. Quick spin to get any residual down and aspirate remaining with 200 µl pipette tip.
11. Wash pellet with 1ml of ice cold 80% ethanol (made with Nuclease Free H₂O), centrifuge for 15 minutes at 4°C, 12,200 RCF.
12. Aspirate off the aqueous with a 1 ml pipette tip. Quick spin to get any residual down and aspirate remaining with 200 µl pipette tip.
13. Open tube and air-dry pellet for 2-3 minutes.
14. Resuspend pellet in 30 µl Ambion Nuclease free H₂O and keep on ice.

15. Add reagents for DNase I digestion, pipette gently up/down to mix after final addition of enzyme.
 - 30 μ l RNA pellet dissolved in Ambion Nuclease free H₂O
 - 3.7 μ l DNase 10X Buffer
 - 2 μ l SUPERase•In
 - 2 μ l Ambion DNase1
 - 37.7 μ l final volume
16. Incubate at 37°C for 30 minutes in a heat block, then store at -80°C until can proceed with subsequent column purification, or proceed directly to column purification.
17. If samples are frozen, Thaw DNase-treated samples on ice.
18. While thawing make up RLT/BME solution:
 - 99 μ l RLT
 - 1 μ l BME
19. Add 100 μ l of RLT/BME to the DNase sample and pipette up/down well to mix.
20. Add 100 μ l 70% ethanol (made with Nuclease Free H₂O) and mix well.
21. Transfer to a Qiacolumn and place into a collection tube. Let bind for 5 minutes at RT room temperature.
 - a. Unless otherwise noted - following steps all done at Room Temperature
22. Spin 5 minutes at 2,655 RCF
23. Collect flow-through and reapply to column. Let bind for 5 minutes.
24. Spin 1 minute at 6,797 RCF.
25. Place column into a new collection tube and add 700 μ l RW1 to the column.
26. Spin 30 sec at 6,797 RCF.
27. Place column into a new collection tube and add 500 μ l RPE to the column.
28. Spin 30 sec at 6,797 RCF.
29. Place column into a new collection tube and add 500 μ l RPE (2nd wash) to the column.
30. Spin 30 sec at 6,797 RCF.
31. Place column into a new collection tube and add 500 μ l 80% ethanol (made with Nuclease Free H₂O) to the column.
32. Spin 2 minutes at 6,797 RCF.
33. Place column into a new collection tube and spin 5 minutes at 6,797 RCF with the cap open; if the cap closes during the spin, give it another 2 minute spin at 6,797 RCF.
 - a. Twist caps to the side to prevent them from closing during the spin
34. Place column in 1.5 ml RNase-free eppendorf tube (from Qiagen Kit) and add 30 μ l RNase-free H₂O (from Qiagen kit) to the column. Let sit for 2 minutes.
35. Spin 5 minutes at 2,655 RCF and then 30 seconds at 10,621 RCF.
36. Add another 30 μ l RNase-free water and let sit for 2 minutes. (on Ice)
37. Spin 5 minutes at 2,655 RCF and then 30 seconds at 10,621 RCF.
38. Measure final volume: it should be equal to input volume less column retention (2-4 μ l)
 - a. Keep RNA on Ice from this point forward
39. Aliquot amounts needed for QC/QA
 - a. Nanodrop or Ribogreen assay to check concentration
 - b. Bioanalyzer 'pico' chip to check integrity
40. Store RNA samples at -80°C until ready to be used.